

ISTVT: Interpretable Spatial-Temporal Video Transformer

EE656 Course Project: Group 2

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Objectives of the ISTVT Framework

- Develop a robust video-based Deepfake detection framework by jointly learning spatial artifacts and temporal inconsistencies.
- Improve detection performance and generalization through decomposed spatial-temporal self-attention and the self-subtract mechanism.
- Enhance model transparency by providing interpretable spatial and temporal attention visualizations to explain prediction decisions.

Limitations of Existing Deepfake Detection Methods

Frame-Based Approaches

- Detect Deepfakes using single-frame spatial artifacts such as blurred boundaries and inconsistent textures.
- **Limitations**
 - Unable to exploit temporal inconsistencies across frames.
 - Performance degrades on high-quality and compressed Deepfake videos.
 - Limited cross-dataset generalization due to overfitting to specific manipulation methods.

Video-Based Approaches

- Incorporate temporal information using LSTMs, 3D CNNs, or conventional video models.
- **Limitations**
 - Fail to effectively model subtle spatial-temporal inconsistencies.
 - Higher computational complexity with limited performance gains.
 - Lack interpretability, making prediction decisions difficult to explain.

ISTVT Architecture: Core Components

Feature Extraction

- The original ISTVT employs Xception Entry Flow to capture low-level facial texture features.
- Our implementation replaces **Xception** with a pretrained **EfficientNet-B0**, providing stronger feature representations and faster convergence.

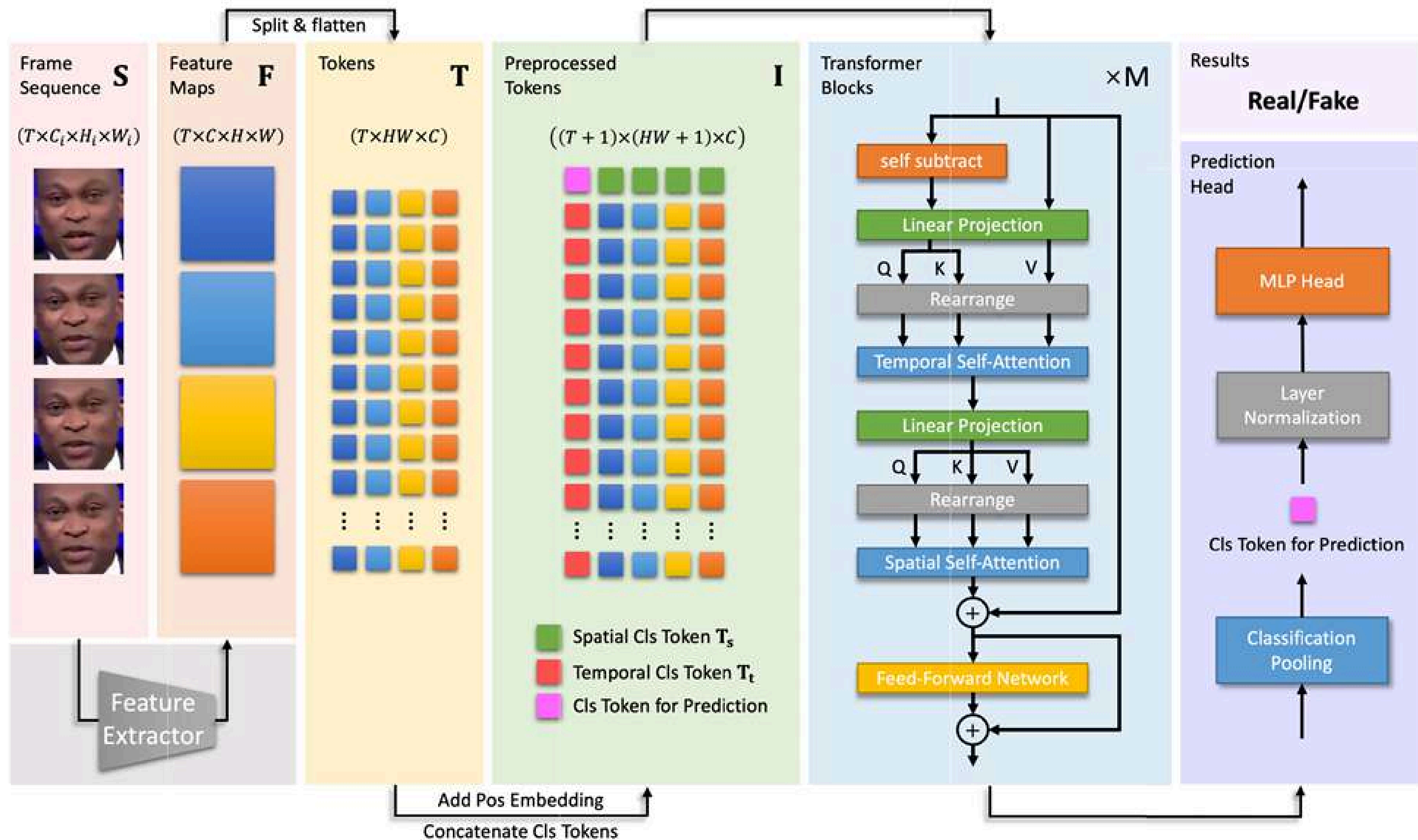
Spatial-Temporal Transformer

- Learns complementary information through decomposed self-attention:
 - **Spatial Attention**: Detects within-frame artifacts (blurred boundaries, texture inconsistencies).
 - **Temporal Attention**: Captures inter-frame inconsistencies caused by independent frame generation.

Prediction Head

- Aggregates learned spatial-temporal representations.
- Performs binary classification (Real v/s Fake) using an **MLP classifier**.

Model Architecture



A. EfficientNet-B0 Feature Extractor (Implementation Modification)

- **Our Implementation:** Replaced **Xception** with pretrained **EfficientNet-B0**
- **Input:** Video frame sequence $S \in \mathbb{R}^{T \times C_i \times H_i \times W_i}$
- EfficientNet-B0 extracts hierarchical facial texture features from each frame using:
 - MBConv (Mobile Inverted Bottleneck) blocks
 - Depthwise Separable Convolutions (reducing computation)
 - Squeeze-and-Excitation (SE) blocks (emphasizing important feature channels)
- **Output:** Feature maps $F \in \mathbb{R}^{T \times C \times H \times W}$
- F is split into 1×1 patches and flattened to get tokens: $T \in \mathbb{R}^{T \times HW \times C}$
- Tokens are concatenated with **Spatial Classification Token** $T_s \in \mathbb{R}^{T \times 1 \times C}$ & **Temporal Classification Token** $T_t \in \mathbb{R}^{1 \times (HW+1) \times C}$
- Position embeddings are added to form: $I \in \mathbb{R}^{(T+1) \times (HW+1) \times C}$
- I is passed through M spatial-temporal transformer blocks.
- The final classification token is processed by the MLP head to predict Real/Fake.
- **Loss Function: Binary Cross-Entropy (BCE).**

B. Decomposed Spatial-Temporal Transformer Block

- **Input to Transformer:** Feature maps extracted using EfficientNet-B0 are flattened into tokens, concatenated with Spatial CLS and Temporal CLS tokens, and enriched with positional embeddings before entering the transformer.
- **Linear Projection:** The input tokens are projected into Queries (Q), Keys (K), and Values (V), and split into multiple attention heads.

$$O_t(:, j, :, :) = \text{softmax} \left(\frac{Q(:, j, :, :) K(:, j, :, :)^T}{\sqrt{D}} \right) V(:, j, :, :)$$

- **Temporal Self-Attention:** Computes attention across frames while preserving the same spatial location.

$$O_s(k, :, :, :) = \text{softmax} \left(\frac{Q(k, :, :, :) K(k, :, :, :)^T}{\sqrt{D}} \right) V(k, :, :, :)$$

- **Spatial Self-Attention:** Computes attention across spatial tokens within each frame.

$$O_s(k, :, :, :) = \text{softmax} \left(\frac{Q(k, :, :, :) K(k, :, :, :)^T}{\sqrt{D}} \right) V(k, :, :, :)$$

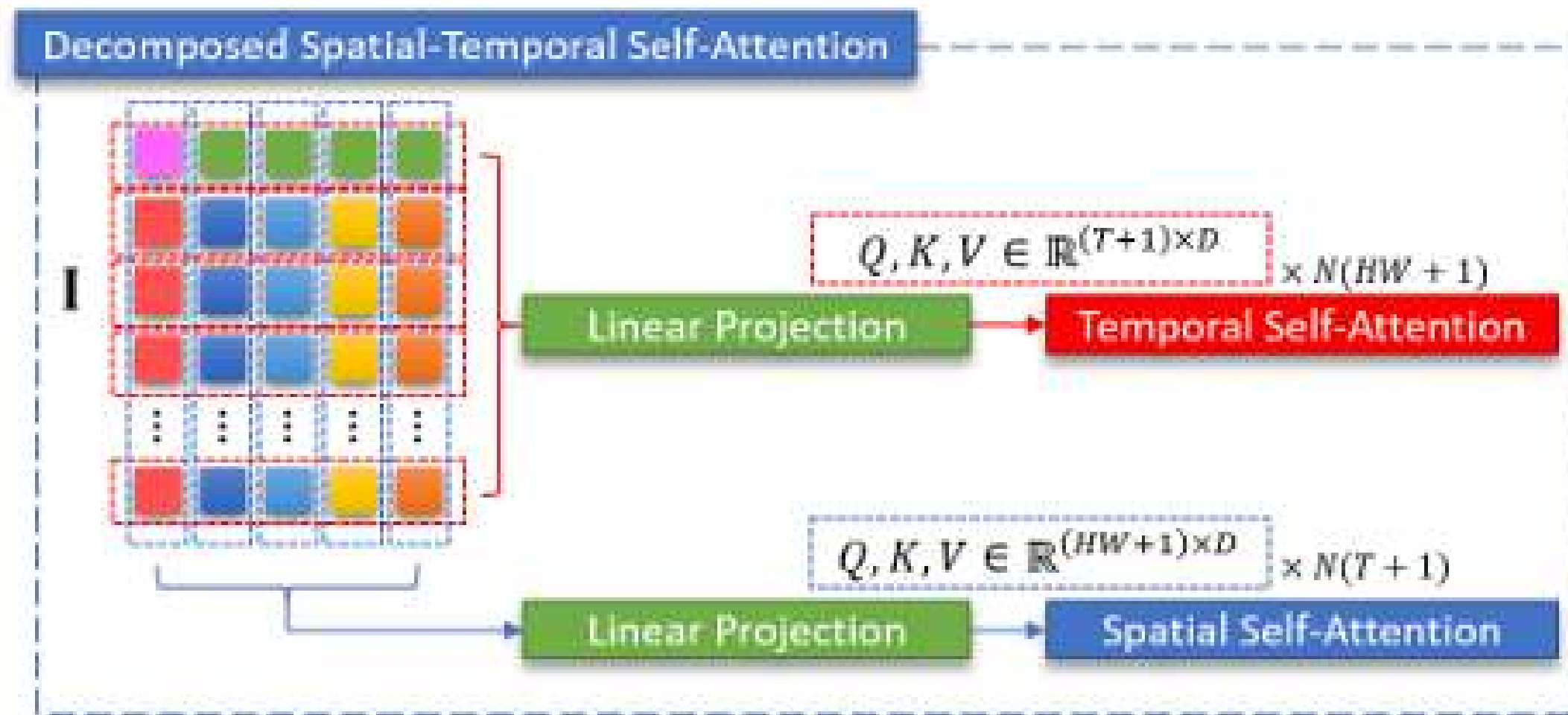
- **Efficient Implementation:**

Temporal: Reshape to $N \times (HW + 1) \times (T + 1) \times D$

Spatial: Reshape to $N \times (T + 1) \times (HW + 1) \times D$

- **Complexity Reduction:**

From $O(T^2 H^2 W^2)$ to $O(T^2 + H^2 W^2)$



C. Self-Subtract Mechanism

- **Objective:** Suppress redundant temporal information and emphasize inter-frame inconsistencies.
- **Input:** $I \in \mathbb{R}^{(T+1) \times (HW+1) \times C}$
- **Self-Subtract Operation:** $I' = \text{cat} \left(I(0:2, :, :), I(2 :, :, :) - I(1 : -1, :, :) \right)$
- **Feature Projection:**
 - Queries (Q) and Keys (K) are computed from the enhanced features I'
 - Values (V) are computed from the original features I .
- **Key Characteristics:**
 - Preserves the classification tokens.
 - Enhances discriminative temporal changes while suppressing static information.
- **Benefits:**
 - Improves sensitivity to frame-to-frame inconsistencies.
 - Enhances robustness and Deepfake detection performance.

D. Proposed Modifications & Implementation

- **Pretrained Texture Backbone:**
 - Replaced the original from-scratch **Xception** network with a pretrained CNN backbone (**EfficientNet-B0 via timm**). This transfer-learning approach extracts better low-level forgery textures given the limited dataset size.
- **Face Cropping Preprocessing:**
 - Utilized OpenCV's DNN face detector (**SSD + ResNet-10**) to crop and center facial regions, ensuring the model learns facial artifacts rather than background noise.
- **Consecutive-Frame Sampling:**
 - Extracted overlapping **6-frame clips** per video (sliding window with stride 4) to accurately capture true frame-to-frame short-term inconsistencies for the self-subtract mechanism.

E. Dataset and Training Setup

- **Dataset:** Evaluated on Celeb-DF-v2, strictly balanced to 590 real and 590 fake videos.
- **Data Splitting:** Applied a strict Video-Level 80/20 train/validation split to prevent frame leakage between sets.
- **Optimization Strategy:**
 - Implemented Layer-wise learning rates (Backbone LR: $1e-5$, Head LR: $3e-4$) to preserve pretrained features while rapidly training the transformer head.
 - Utilized a linear warmup (2 epochs) followed by cosine learning rate decay.
 - Trained for 20 epochs using Mixed Precision (AMP) for optimal T4 GPU memory and speed.

Results & Performance

- **Clip-Level Metrics:**
 - Accuracy: 93.46%.
 - AUC: 0.9897
- **Video-Level Aggregated Metrics** (Mean of Clip Probabilities):
 - Accuracy: 95.76%
 - AUC: 0.9959
- **Video-Level Confusion Matrix:**
 - True Negatives: 110 | False Positives: 8
 - False Negatives: 2 | True Positives: 116
- **Takeaway:** The model demonstrates exceptionally **high discrimination capability on Celeb-DF-v2**, reliably detecting both real and synthesized sequences.

Interpretability by Visualization

- **Relevance Propagation** (Algorithm 1):
 - Implemented to **extract gradients** and generate interpretable spatial and temporal heatmaps.
- **Spatial Heatmaps:**
 - Effectively isolate frame-level manipulation artifacts, such as **spatial blending** and **structural distortions**.
- **Temporal Heatmaps:**
 - Highlight inter-frame inconsistencies and temporal flickering across the 6-frame sequence.
- **Visualization:**
 - Heatmaps are mapped to color arrays (**JET colormap**) and overlaid onto original RGB frames with **50% opacity**, providing clear, human-readable explanations of the model's decision-making process.

Interpretability by Visualization

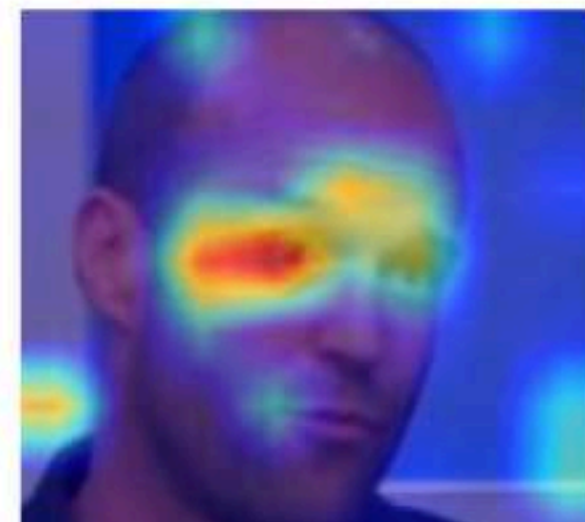
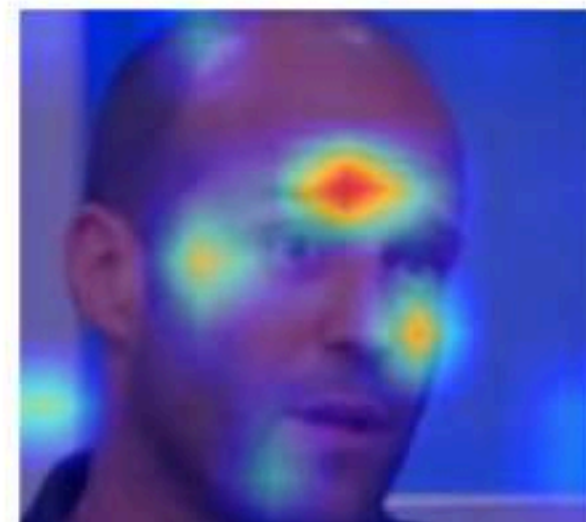
Original Clip



Spatial Blending



Temporal Flickering



Conclusion and Key Takeaways

- **Successful Reproduction:**

- Accurately implemented the core ISTVT contributions, including decomposed spatial-temporal self-attention and the self-subtract mechanism.

- **Impact of Modifications:**

- Demonstrates that utilizing a pretrained backbone with transfer learning is highly effective for training complex video-transformers in low-data environments.

- **State-of-the-Art Performance:**

- Achieved an outstanding 0.9959 Video-Level AUC on the Celeb-DF-v2 dataset.

- **Transparency:**

- The generated spatial and temporal relevance maps validate that the model correctly learns to detect deepfakes based on logical forgery clues rather than background bias.

Thank You

References:

- [1] ISTVT: Interpretable Spatio-Temporal Video Transformer for Deepfake Detection
- [2] OpenCV DNN Face Detector & timm PyTorch Image Models
- [3] Celeb-DF-v2 Dataset